

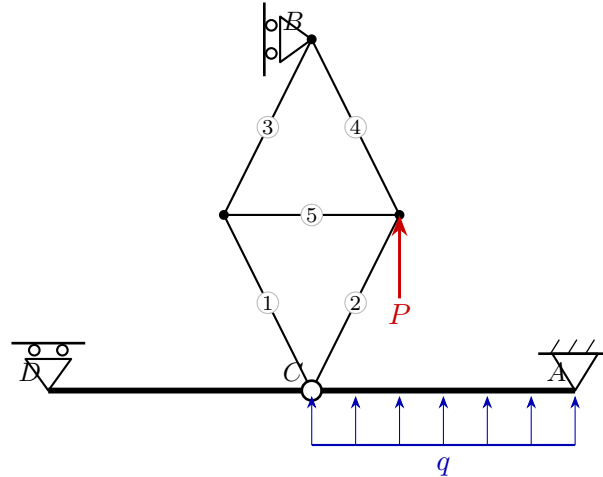
# University of Natural Resources and Life Sciences, Vienna (BOKU)

## VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

Date: 26.06.2026

### Model solution



#### 1. Static determinacy

Counting criterion  $n = 3k - (a + z)$  (bodies  $k$ , reactions  $a$ , interaction forces  $z$ ):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

#### 2. Support reactions and hinge forces

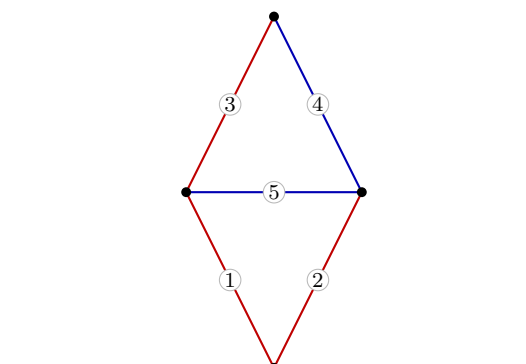
$$A: R_x = -3.75, R_y = -12 \quad D: R_x = 0, R_y = -9 \quad B: R_x = 3.75, R_y = 0$$

$$C: H_x = 3.75, H_y = 15$$

#### 3. Truss member forces

| Member | Force $S$ [kN] | Type        |
|--------|----------------|-------------|
| 1      | 4.19           | tension     |
| 2      | 12.58          | tension     |
| 3      | 4.19           | tension     |
| 4      | -4.19          | compression |
| 5      | -3.75          | compression |

red: tension (+)   blue: compression (-)



**4. Section-force diagrams of the beams**